

Project Design Phase

Solution Architecture

Date	15 July 2026
Team ID	XXXX
Project Name	Emotion Detection and Learning Support Engine
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process - with many sub-processes - that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Solution Architecture Diagram:

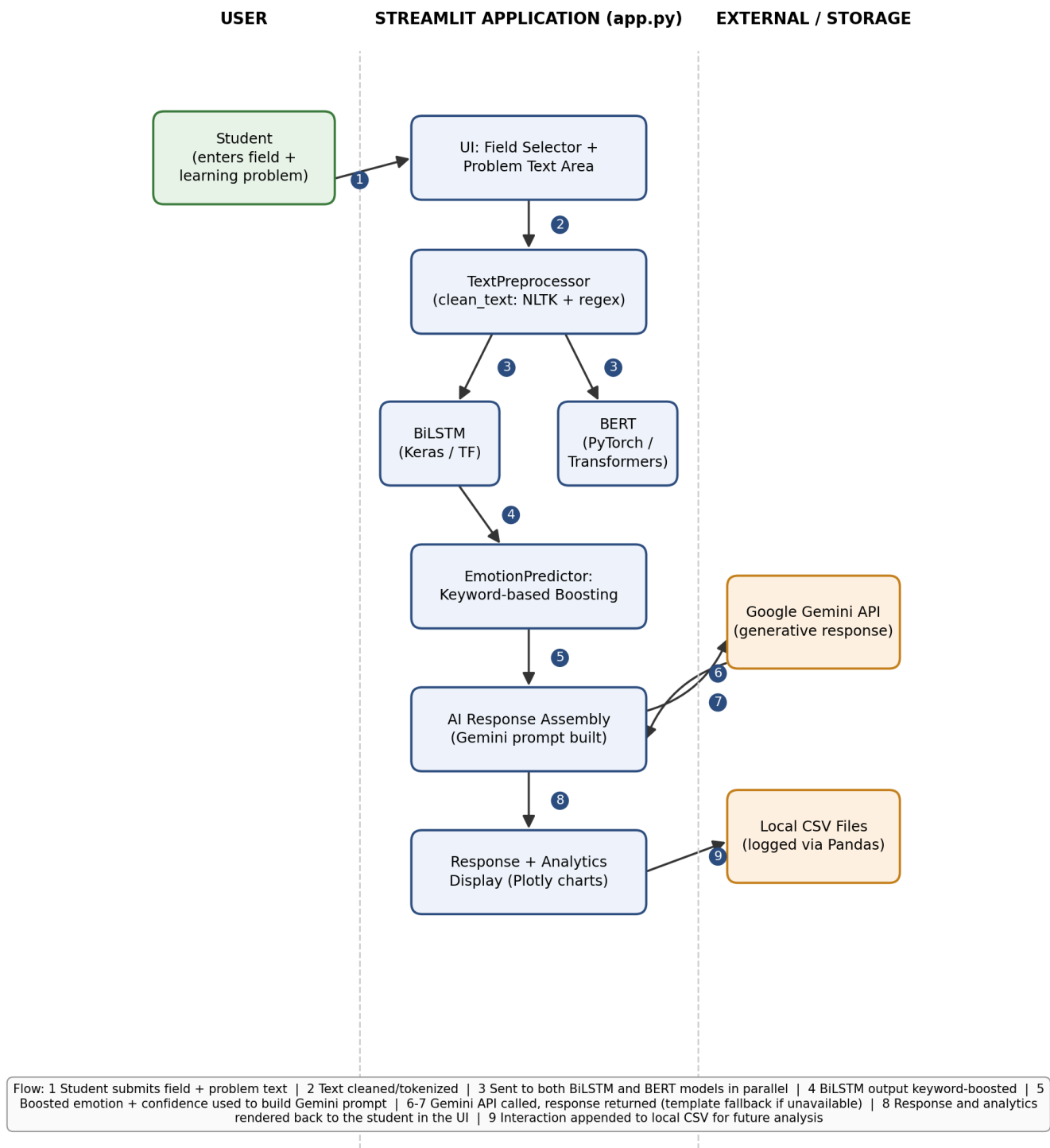


Figure 1: Architecture and data flow of the Emotion Detection and Learning Support Engine, showing how a student's text input flows through preprocessing, dual-model emotion classification (BiLSTM and BERT), keyword-based boosting, Gemini-powered response generation, and interaction logging.